PRACTICAL STUDY • 800 VDC AI DATA CENTERS

# DC Arc Flash

## Learnings from simulation, laboratory validation, and standards gaps

**Tony Landry**
Power Systems Competency Center — NAM Leader

**Stuart Sheehan**
CTO Office — Systems Engineer

SCHNEIDER ELECTRIC

# DATA CENTER FACILITY - POWER DISTRIBUTION Sub-Project



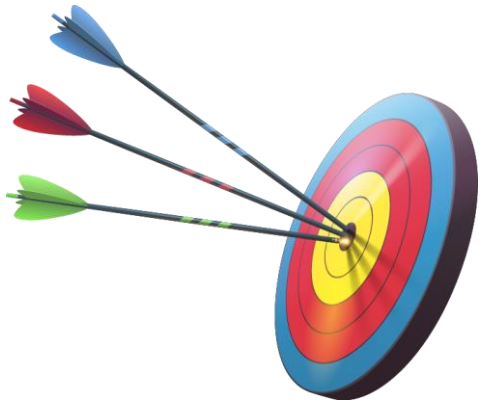


General

CONTEXT

# The study answers one practical question

## How should arc-flash risk be assessed when an active converter shapes the DC fault current?



**01 Frame the gap**

Accepted AC methods do not directly resolve active-converter DC behavior (e.g transient current, limited current).

**02 Compare architectures**

Power-rack and centralized systems produce different transient signatures.

**03 Connect model to evidence**

Simulation establishes the baseline; the validation plan tests waveform, energy, and protection response.

**04 Translate to action**

Product data, protection testing, and standards must evolve together.

# Today’s accepted methods were built around AC systems




The objective is consistent:

**quantify risk and select appropriate PPE.**

1. **PPE tables**
   NFPA 70E Table 130.7
2. **Incident-energy calculation**
   IEEE 1584
3. **Laboratory testing**
   UL RP 2986

**Common limitation**

**All three methods are rooted in AC-system assumptions**

METHODS

# Active converters expose a gap in DC guidance

## WHAT EXISTS

### NFPA 70E PPE selection

Table 130.7(C)(15)(b) Arc Flash PPE Categories for dc Systems

| Equipment | Arc Flash PPE Category | Arc Flash Boundary |
|---|---|---|
| Storage batteries, dc switchboards, and other dc supply sources | | |
| Parameters: Greater than 150 volts and less than or equal to 600 volts | | |
| Maximum arc duration and minimum working distance: 2 sec @ 455 mm (18 in.) | | |
| Available fault current less than 1.5 kA | 2 | 900 mm (3 ft) |
| Available fault current greater than or equal to 1.5 kA and less than 3 kA | 2 | 1.2 m (4 ft) |
| Available fault current greater than or equal to 3 kA and less than 7 kA | 3 | 1.8 m (6 ft.) |
| Available fault current greater than or equal to 7 kA and less than 10 kA | 4 | 2.5 m (8 ft) |

### Annex D.5 calculation methods

$$I_{arc} = 0.5 \times I_{bf}$$
$$IE_n = 0.01 \times V_{sys} \times I_{arc} \times T_{arc} / D^2$$

where:

$I_{arc}$ = arcing current, amperes

$I_{bf}$ = system bolted fault current, amperes

$IE_n$ = estimated dc arc flash incident energy at the maximum power point, cal/cm$^2$

$V_{sys}$ = system voltage, volts

$T_{arc}$ = arcing time, sec

$D$ = working distance, cm

## WHERE IT BREAKS

01 **Developed around PV or battery-type sources**

02 **Known and proven to be conservative**

03 **Recognized, but not standardized**

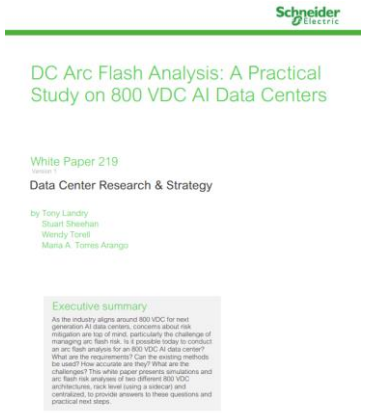
04 **Does not cover active-converter DC systems as expected in 800 Vdc AI Data Center**

STUDY

# A transient workflow connects simulation to laboratory validation

## Recent transient-current research

Two papers presented at the IEEE IAS Electrical Safety Workshop in 2025 and 2026.


Schneider Electric

DC Arc Flash Analysis: A Practical Study on 800 VDC AI Data Centers

White Paper 219

Data Center Research & Strategy

by Tony Landry
Stuart Sheehan
Wendy Torell
Maria A. Torres Arango

Executive summary

As the industry aligns around 800 VDC for next generation AI data centers, concerns about risk mitigation are top of mind, particularly the challenge of managing arc flash risk. Is it possible today to conduct an arc flash analysis for an 800 VDC AI data center? What are the requirements? Can the existing methods be used? How accurate are they? What are the challenges? This white paper presents simulations and arc flash risk analyses of two different 800 VDC architectures, rack level (using a sidecar) and centralized, to provide answers to these questions and practical next steps.

## Practical study baseline

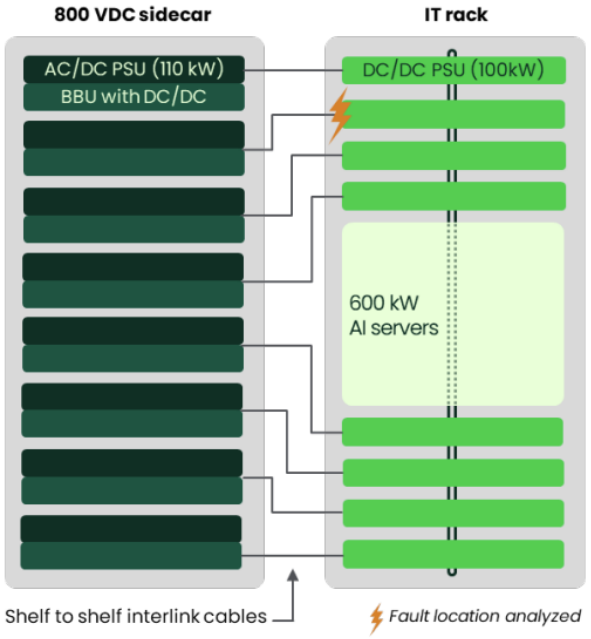
The source deck references an August 2026 whitepaper covering the modeled architectures and assumptions.

| | | |
|---|---|---|
| 1 | **Model** | Simulate converter-shaped short-circuit waveforms. |
| 2 | **Calculate** | Translate transient current into incident-energy estimates. |
| 3 | **Validate** | Replicate the modeled system in a laboratory environment. |
| 4 | **Compare** | Reconcile measured waveforms, energy, and protection response. |

ARCHITECTURES

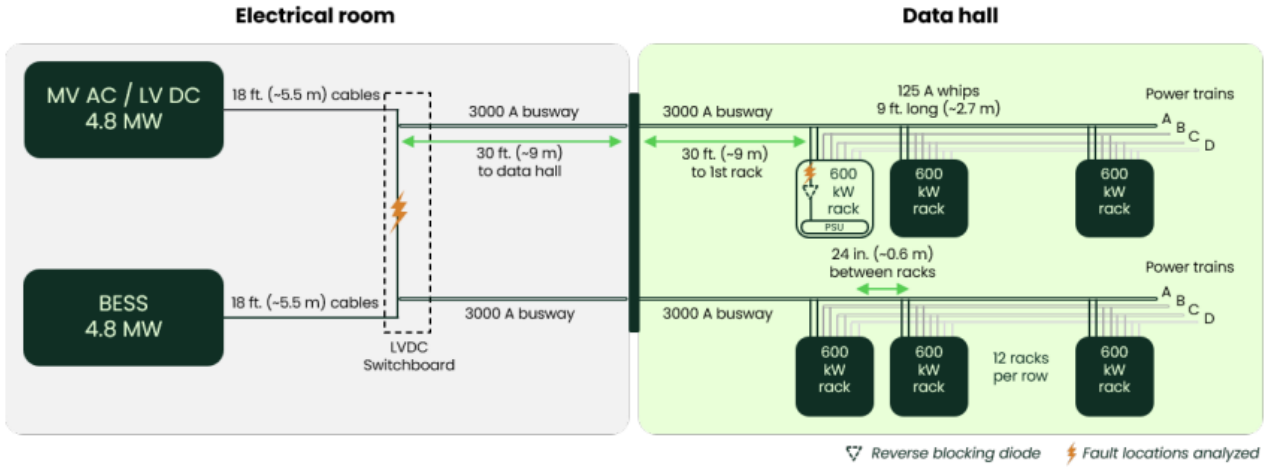
# Two architectures frame the comparison

## POWER RACK




Distributed power conversion at the rack level

## CENTRALIZED



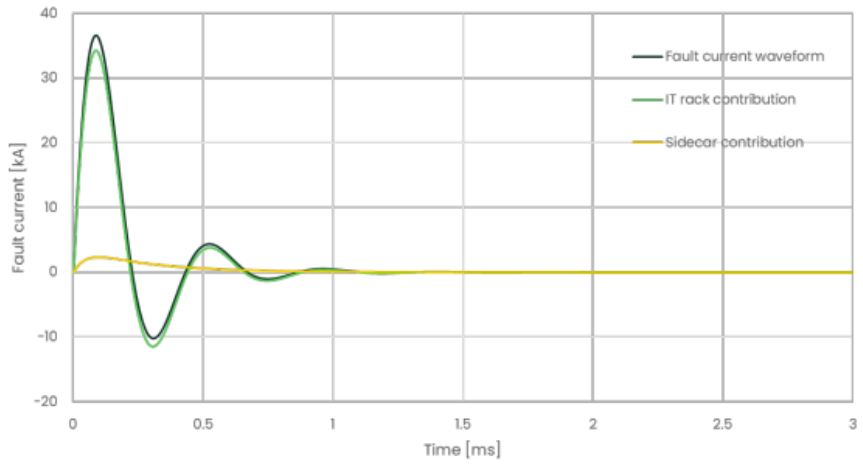

Centralized distribution creates a different source path and fault-current envelope.

**Same decision question. Different transient behavior.**

SIMULATION

# Architecture changes the transient signature

## POWER RACK

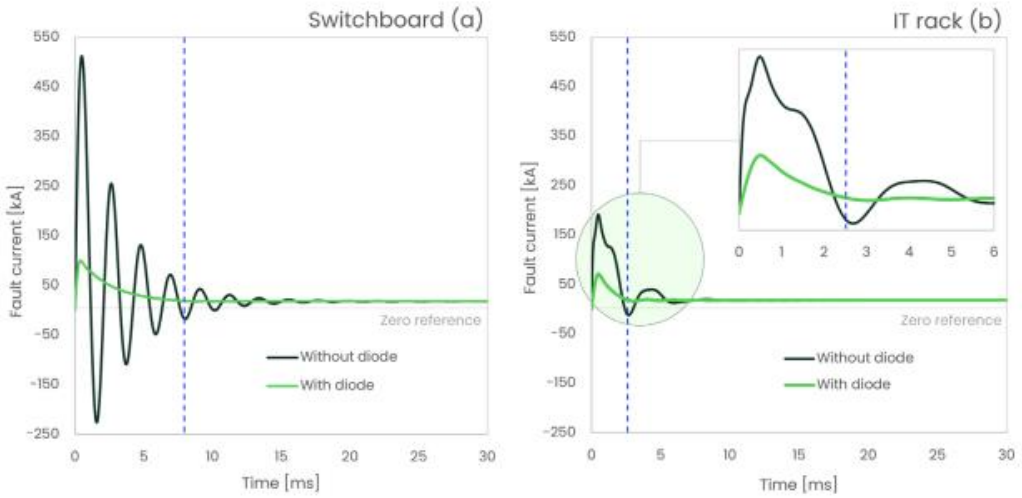



**Simulated short-circuit waveform**

| Annex R incident energy | ETAP incident energy | Percent difference ETAP vs. Annex R |
|---|---|---|
| 0.05 cal/cm² | 0.01 cal/cm² | ~ -80% |

*Etap simulation based on Stoke and Oppenlander equations*

## CENTRALIZED

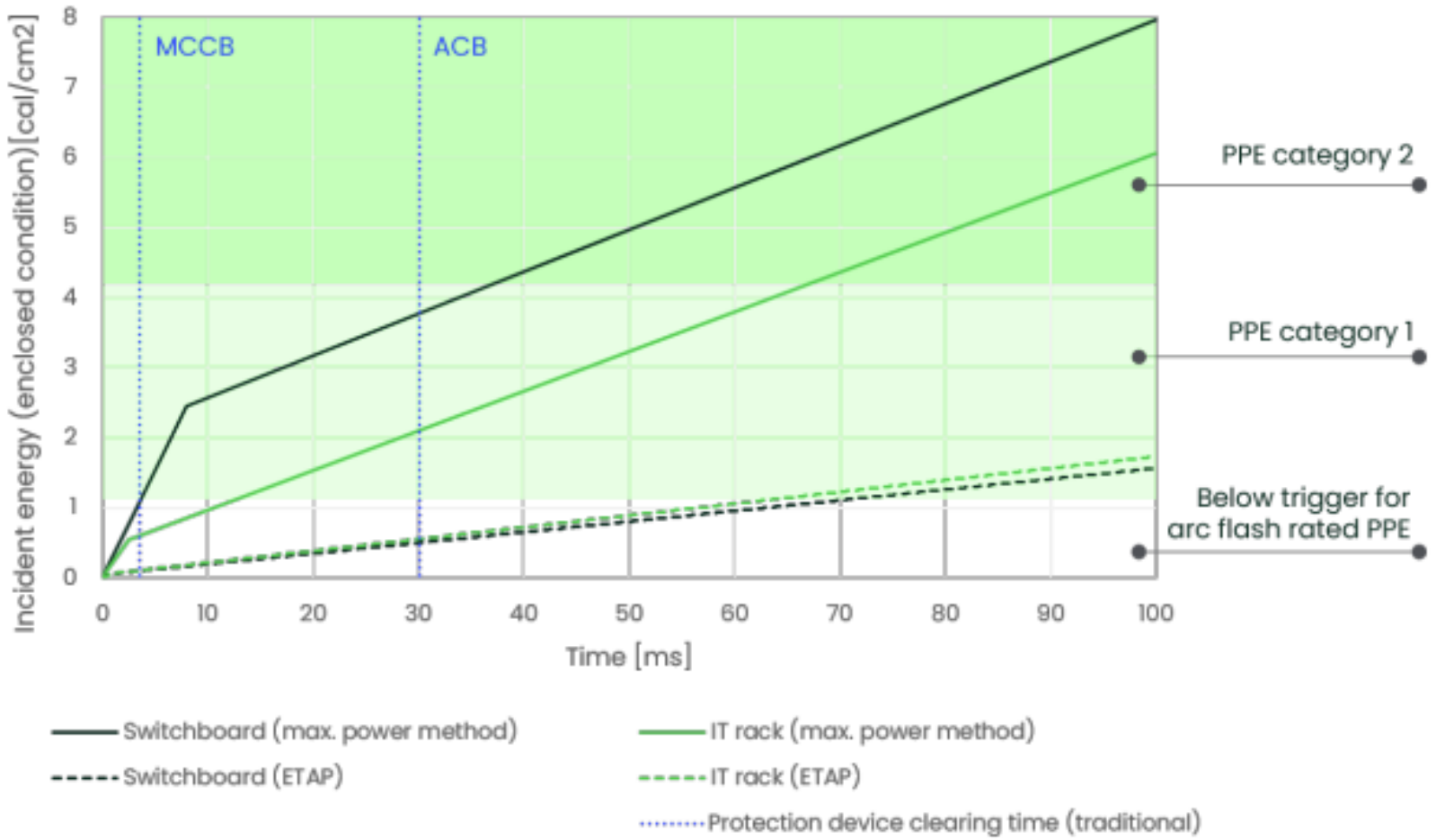



**Simulated short-circuit waveform**

**Incident energy is architecture-specific because current magnitude and duration evolve differently.**

SIMULATION

# PPE classification depends on current and clearing time




HOW TO READ IT

**Incident energy increases over time.**

**Faster protection clearing reduces exposure.**

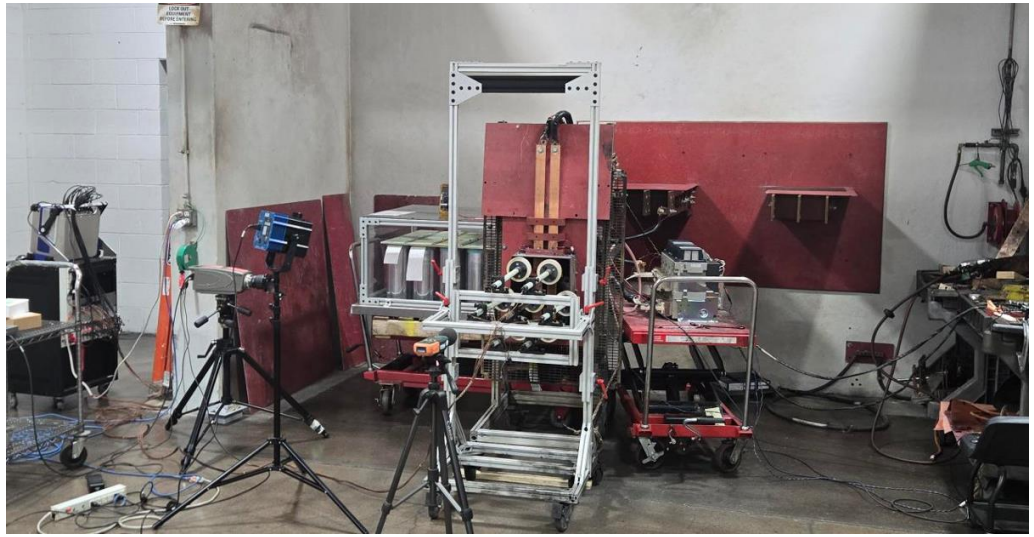
**Converter current limiting feature behavior shift the operating point.**

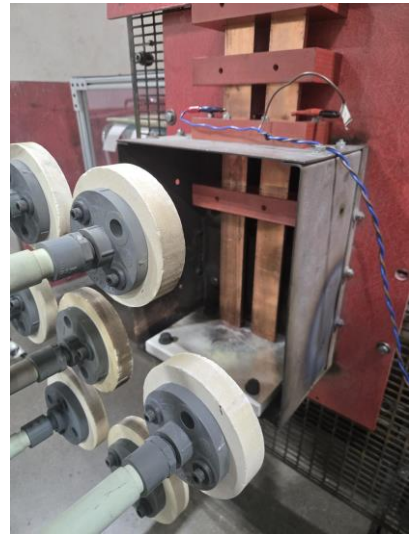
VALIDATION

# Validation mirrors the modeled systems

**The laboratory setup is designed to reproduce the modeled architectures and capture the converter-shaped fault response.**

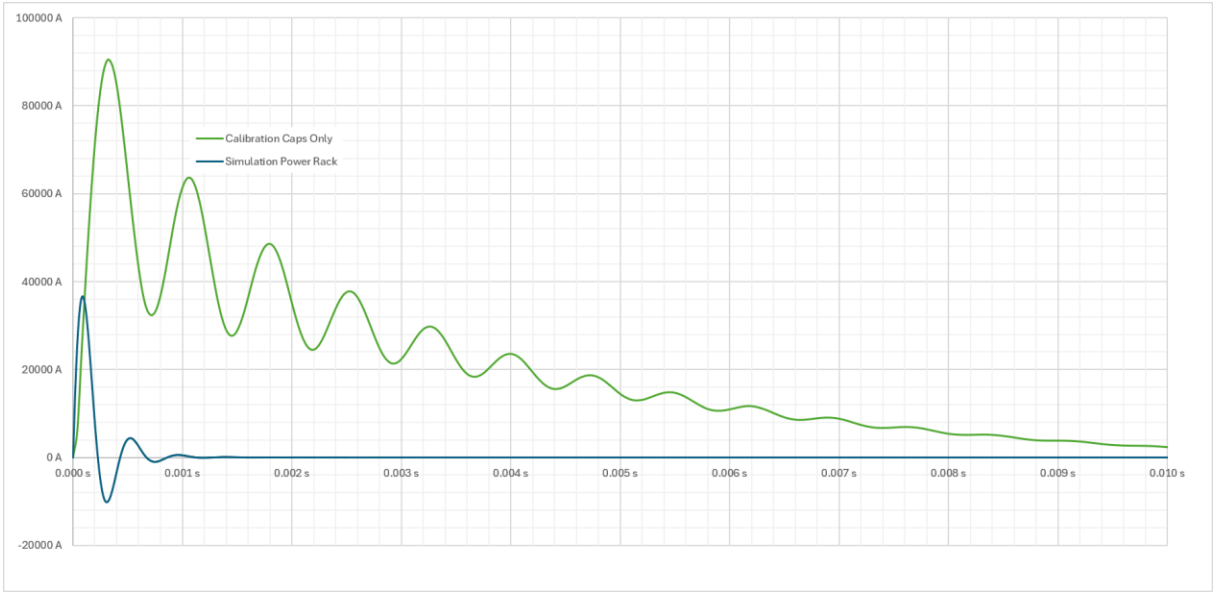
Planned evidence
- instrumented lab setup
- synchronized current and voltage capture
- architecture-specific protection response



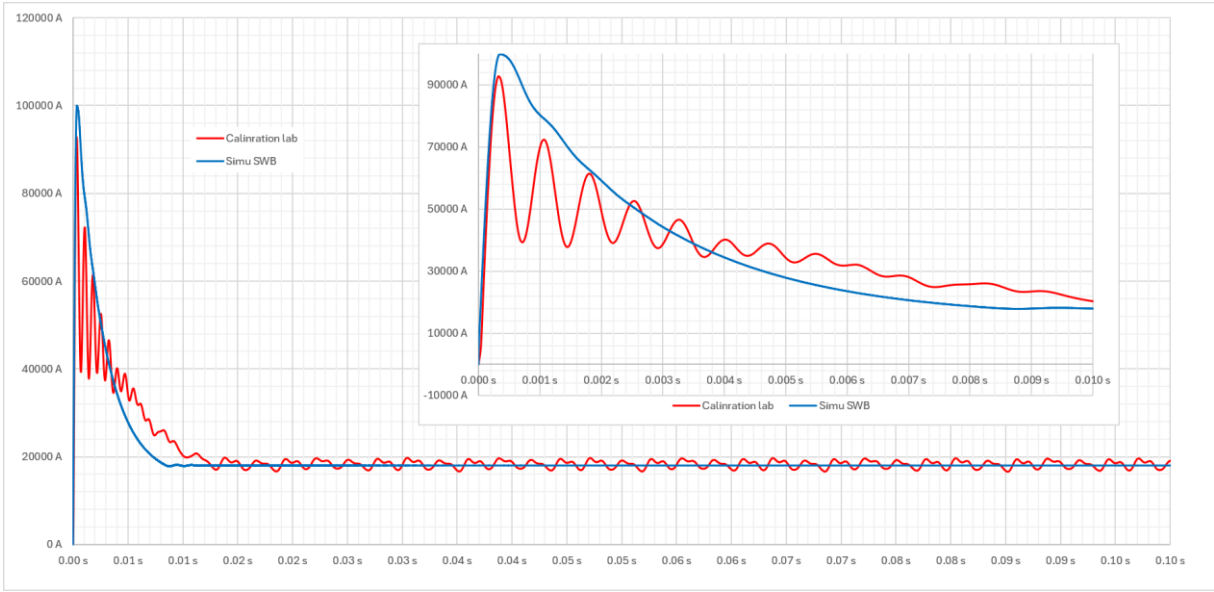


Tests performed September 2026 at SE lab – Results shared with the Fire Protection Research Fundation Porject on DC arc flash

# Laboratory current waveforms vs Simulation




**Power-rack baseline**
**(Was test with more energy than simulated)**



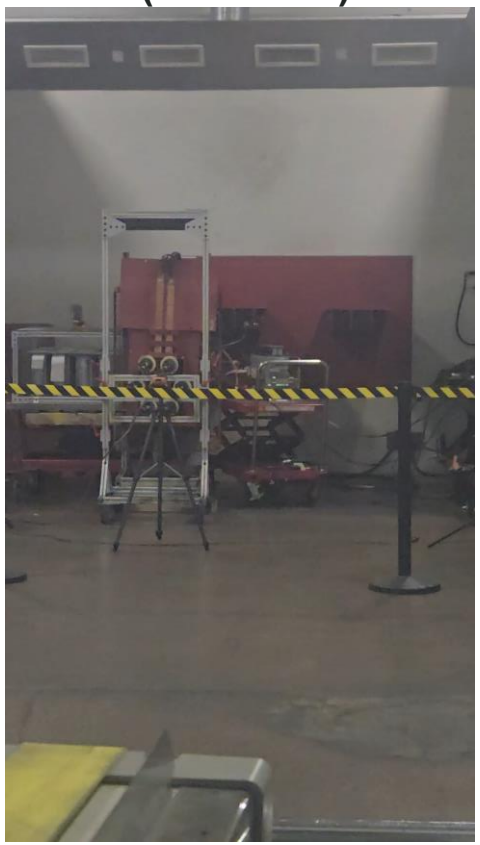

**Centralized baseline**
**(@ Switchboard location)**

# Pictures and Videos of arcing tests

**When Self Extinguish (<0.1 cal/cm²)**



**When Sustain (2 cal/cm²)**



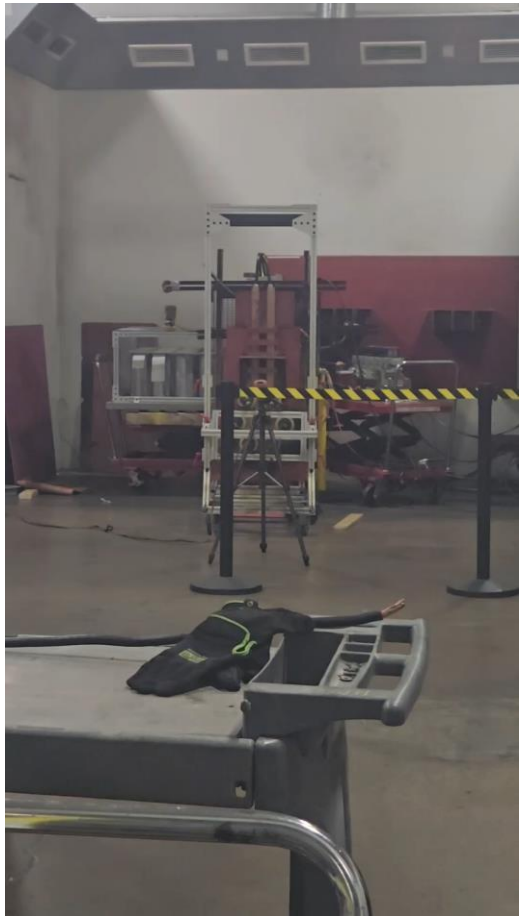
| | | |
|---|---|---|
| <0.1 cal/cm² | 21 tests | 54% |
| <0.2 cal/cm² | 7 tests | 18% |
| <1.2 cal/cm² | 6 tests | 15% |
| <4 cal/cm² | 3 tests | 8% |
| >4 cal/cm² | 2 tests | 5% |

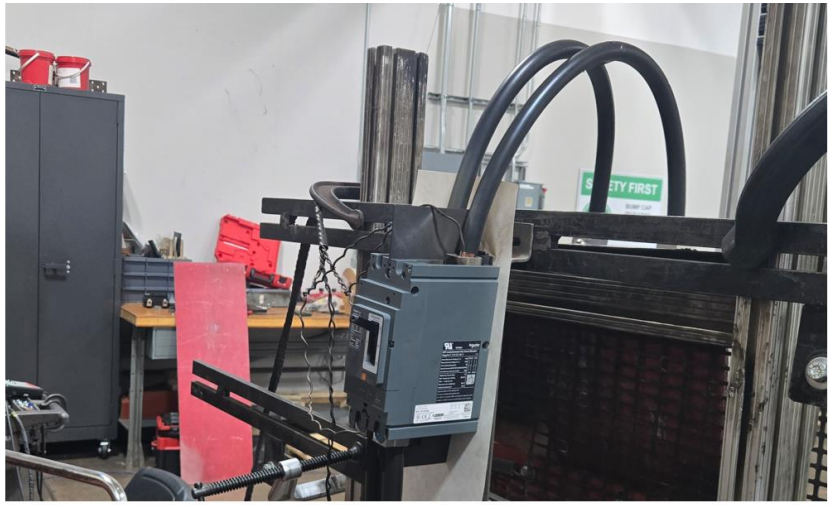
39 tests total

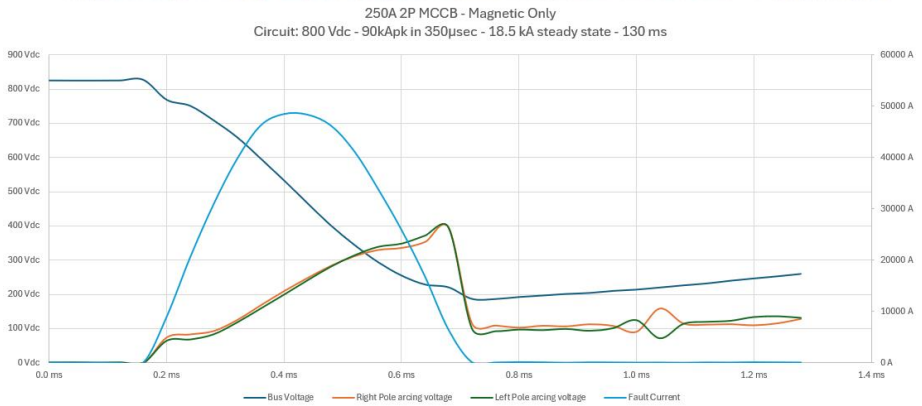
**80% of test, arc did not sustain**

# Test with 250 A MCCB

**250 A MCCB**

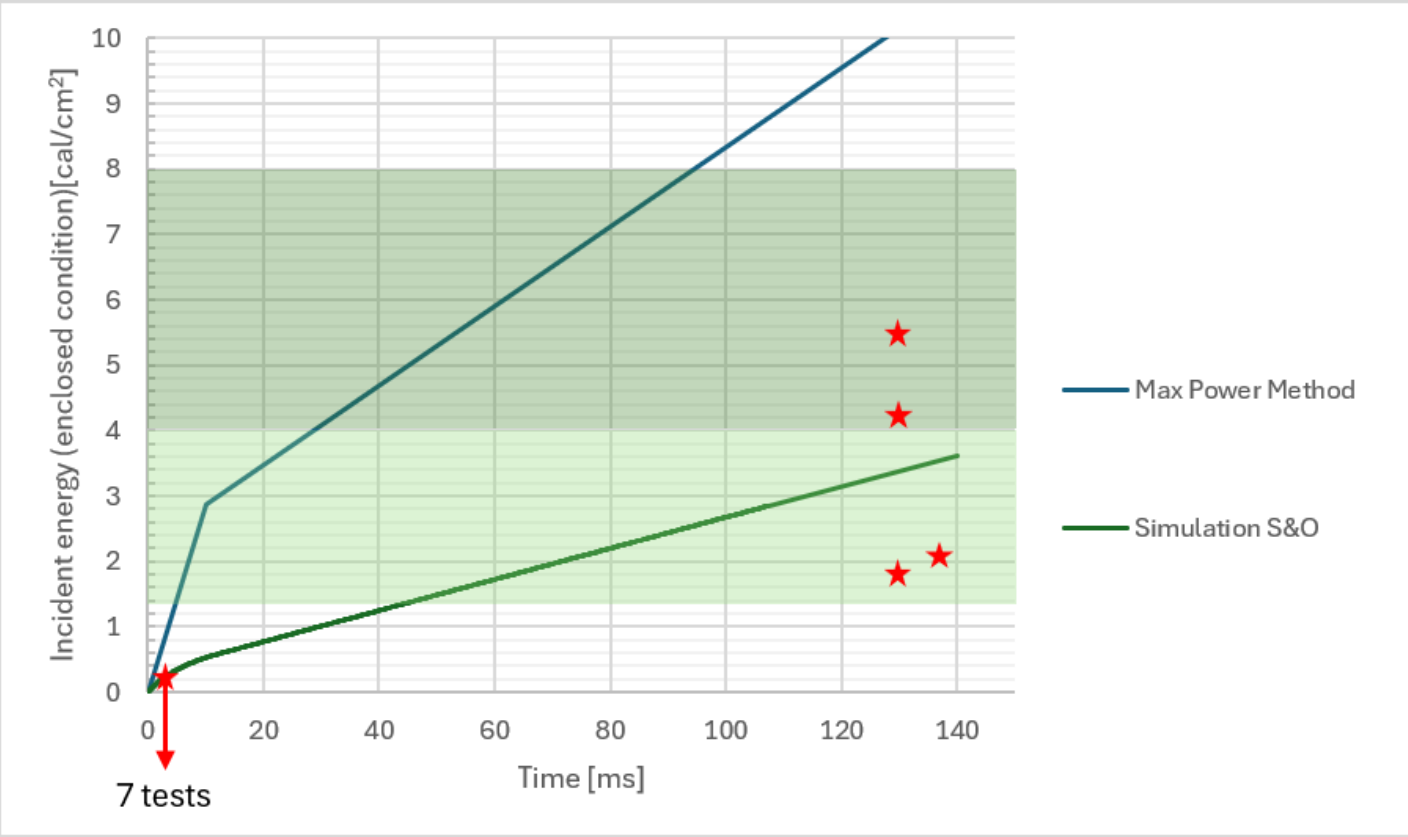







**Clearing time: 0.6 ms**
**IE= 0.03 cal/cm²**

# Comparison Simulation vs Test Results




Tests without overcurrent protection / Current duration set at ≈130 ms by test station

Comparison on 11 tests (similar configuration)

- 7 tests < 0.1 cal/cm²
  → arc self extinguished
- 2 tests ≈ 2 cal/cm²
  → Arc sustained between busbar
- 2 tests > 4 cal/cm²
  → Arc sustained between busbars and enclosure, it means there was « 2 arcs »

# Three key assumptions dominate the result

## 01

### Converter Current limiting function

Rapid shutdown and current magnitude / duration limits must be modeled explicitly.

## 02

### Protection clearing time

Use the converter-specific fault waveform; legacy breaker or fuse curves may not apply as-is.

## 03

### Reverse current blocking feature

A rack can contribute more fault current than the upstream source; blocking can reduce hazard and improve coordination.

**Assumptions are not footnotes — they determine prospective current, exposure duration, and coordination.**

IMPLICATIONS

# Product data and standards must evolve together

PRODUCT AND PROTECTION DATA

## Converters

Publish equivalent output impedance and abnormal-condition behavior. Validate current-limiting and shutdown features.

## Protection devices

Update test methods and ratings to reflect converter-shaped fault behavior.

DC ANALYSIS STANDARDS

## IEEE work in progress

P3002.12 — DC arcing-fault hazard calculation
P3002.14 — DC fault calculations

## Fire Protection Research Foundation Project

Research on DC arc-flash behavior and modeling.
*→ All test results presented will be used by the project to better understand arc behavior*

CALL TO ACTION

# Three priorities move DC arc-flash practice forward

## 01

### Standardize analysis

Advance DC-specific calculation guides and research.

## 02

### Characterize converters

Publish abnormal-condition behavior and equivalent impedance.

## 03

### Validate protection

Test clearing behavior against converter-shaped fault waveforms.

Power Distribution Sub-Project page



General


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GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California
General